

Question 1: Object-Oriented Programming (OOP)**[8 marks]****Introduction: (2 marks)**

Object-Oriented Programming (OOP) is a programming paradigm that structures software design around objects rather than

Core Concepts: (3 marks)

OOP is founded on four core pillars. Encapsulation bundles data and methods within a class and restricts direct external access to internal state using

Examples: (2 marks)

A BankAccount class encapsulates balance (private) and exposes deposit(amount) and withdraw(amount) methods · Encapsulation. SavingsAccount extends

Conclusion: (1 mark)

OOP provides a robust, scalable approach to software

Question 2: Database Normalization**[8 marks]****Introduction: (2 marks)**

Database normalization is a systematic process of organizing a relational database schema to minimize data

Core Concepts: (3 marks)

First Normal Form (1NF): each column must contain atomic (indivisible) values, every row must be unique (primary key)

Examples: (2 marks)

Consider Student_Course(StudentID, StudentName, CourseID, CourseName, InstructorName, Grade). Partial dependency: StudentName depends

Conclusion: (1 mark)

Normalization produces clean, anomaly-free database schemas

Question 3: Operating System Process Management**[8 marks]****Introduction: (2 marks)**

Process management is a core responsibility of the operating system (OS), which must create, schedule, synchronize,

Core Concepts: (3 marks)

Process States: New · Ready · Running · Waiting/Blocked · Terminated. A process moves between

Examples: (2 marks)

Round Robin with quantum=4ms on P1(24ms), P2(3ms), P3(3ms): execution order P1(4),

Conclusion: (1 mark)

Effective process management is the cornerstone of

Question 4: Computer Networks and the OSI Model**[8 marks]****Introduction: (2 marks)**

A computer network is a collection of interconnected devices (hosts, routers, switches) that

Core Concepts: (3 marks)

Layer 7 · Application: provides network services directly to end-user applications. Protocols: HTTP/HTTPS (web), SMTP/IMAP/POP3 (email), FTP/SFTP

Examples: (2 marks)

HTTPS request to www.example.com: DNS resolves domain · IP

Conclusion: (1 mark)

The OSI model is the definitive

Question 5: Data Structures · Binary Trees and BST**[8 marks]****Introduction: (2 marks)**

A tree is a non-linear hierarchical data structure composed of nodes connected by

Core Concepts: (3 marks)

Binary Tree types: Full (every node has 0 or 2 children), Complete (all levels fully filled except

Examples: (2 marks)

BST insertion of [50, 30, 70, 20, 40,

Conclusion: (1 mark)

Binary trees and BSTs provide efficient

Question 6: Sorting Algorithms and Algorithm Analysis**[8 marks]****Introduction: (2 marks)**

Sorting is the process of arranging elements in a defined order (ascending or descending).

Core Concepts: (3 marks)

Bubble Sort: repeatedly swaps adjacent out-of-order elements. Best $O(n)$ (already sorted, with flag),

Examples: (2 marks)

Quick Sort on [3,6,8,10,1,2,1] with pivot=1:

Conclusion: (1 mark)

Choosing the right sorting algorithm

Question 7: Software Engineering · SDLC and Agile**[8 marks]****Introduction: (2 marks)**

Software Engineering is about building software in a systematic way.

Core Concepts: (3 marks)

SDLC has phases like requirements, design, coding, and testing.

Examples: (2 marks)**Conclusion: (1 mark)**

Agile is a popular approach.

Question 8: Computer Architecture · CPU and Memory Hierarchy**[8 marks]****Introduction: (2 marks)**

CPU stands for Central Processing Unit and is the brain of the computer.

Core Concepts: (3 marks)

CPU has ALU, Control Unit and registers. It fetches and executes instructions.

Examples: (2 marks)

Conclusion: (1 mark)

Memory hierarchy includes cache and RAM.

Question 9: Web Technologies · Client-Server Architecture and HTTP**[8 marks]****Introduction: (2 marks)**

Web technologies are used to build websites and web applications.

Core Concepts: (3 marks)

HTTP is the protocol used for web communication. It uses GET and POST methods.

Examples: (2 marks)

Conclusion: (1 mark)

Web security is important to protect data.

Question 10: Cloud Computing · Service Models and Deployment**[8 marks]****Introduction: (2 marks)**

Cloud computing provides resources over the internet on demand.

Core Concepts: (3 marks)

There are three types: IaaS, PaaS and SaaS. AWS and Azure are popular clouds.

Examples: (2 marks)

Conclusion: (1 mark)
